

Data Engineer Interview Questions

1. Data Modeling & Warehousing

What they asked me:

- How do you design a schema for a Data Warehouse?
- Can you explain Slowly Changing Dimensions (SCD Type 2) with an example?

What I said:

For schema design, I explained Star vs Snowflake schema with a Sales Analytics example:

- Fact table: Sales (SalesID, CustomerID, ProductID, Amount, DateID)
- Dimension tables: Customer, Product, Date

For SCD Type 2, I gave a Customer Address Change scenario:

- Old Address row marked as inactive (EndDate + IsCurrent=0)
- New Address row inserted with StartDate & IsCurrent=1

Tips:

Always connect theory to real-world business cases.

Banking (loan applications), Retail (customer orders), or Healthcare (patient records) make strong examples.

2. SQL & Query Optimization

What they asked me:

- Write a SQL query to fetch the top 5 customers by revenue in the last year.
- How do you handle duplicate records?

What I said:

SQL query for top 5 customers:

```
SELECT CustomerId, SUM(Revenue) AS TotalRevenue
```

```
FROM Orders
WHERE OrderDate >= '2024-01-01'
GROUP BY CustomerId
ORDER BY TotalRevenue DESC
LIMIT 5;
```

For duplicates, I used ROW_NUMBER() over partition to keep the latest record:

```
WITH Ranked AS (
  SELECT *, ROW_NUMBER() OVER(PARTITION BY CustomerId
    ORDER BY LastUpdated DESC) AS rn
  FROM Customer
)
SELECT * FROM Ranked WHERE rn = 1;
```

Tips:

Prepare window functions, joins, aggregate queries, and CTEs.

Expect optimization topics: indexes, partitions, and query plans.

3. Big Data (Spark / PySpark)

What they asked me:

- How do you handle skewed data in Spark joins?
- Can you explain the difference between cache, persist, and checkpoint?

What I said:

For skew, I used a 2B row Sales Fact + small Product Dim case:

- Applied salting technique (adding random keys)
- Used broadcast join for small tables

Cache/Persist/Checkpoint:

- Cache: Memory-only
- Persist: Memory + Disk with levels
- Checkpoint: Breaks lineage, saves to HDFS for recovery

Tips:

Show you've faced real Spark issues (e.g., OOM, shuffle failures).

Mention Spark UI for debugging.

4. ETL & Pipelines (ADF / Airflow / Glue)

What they asked me:

- How do you design incremental data loads from SQL Server to Data Lake?
- What's the difference between event-driven vs schedule-driven pipelines?

What I said:

For incremental loads: Use a watermark column (LastModifiedDate), store max value processed, and load only new rows.

Event-driven vs Schedule-driven:

- Event-driven → IoT sensor data triggers pipeline on new file arrival.
- Schedule-driven → Daily batch job at 2 AM.

Tips:

Mention error handling, retries, alerts, monitoring dashboards.

Recruiters value production reliability experience.

5. Cloud & Storage Optimization

What they asked me:

- Compare Delta Lake vs traditional Data Lake.
- How do you optimize queries in Azure Synapse / BigQuery?

What I said:

Delta Lake: ACID transactions, schema enforcement, time travel, merge support.

Data Lake: raw storage, no transaction support.

Synapse/BigQuery tuning:

- Partitioning & Clustering for large datasets

- Materialized Views for pre-aggregations
- Columnstore Indexing for OLAP performance

Tips:

Balance cost efficiency with query speed.

Always add examples: “We saved \$200/month by partition pruning.”

6. Streaming & Real-Time Systems

What they asked me:

- How do you process late arriving Kafka events?
- How would you design a real-time fraud detection system?

What I said:

Kafka late events: Used watermarks + windowing in Spark Structured Streaming.

Fraud detection pipeline:

- Ingest transactions from Kafka
- Apply rule-based filters (transaction > \$10,000 at midnight)
- Feed into ML Model for scoring
- Write suspicious alerts into real-time dashboard (e.g., Power BI, Grafana)

Tips:

Focus on scalability, fault tolerance, and latency.

Recruiters love real-world cases like payments, IoT, or e-commerce.

7. Behavioral & Problem-Solving

What they asked me:

- Tell me about a time when a pipeline failed in production. How did you fix it?
- How do you prioritize performance vs cost in a cloud pipeline?

What I said:

Scenario: Spark job failed due to shuffle OOM.

- Debugged in Spark UI, tuned partitions, enabled adaptive execution.

Performance vs Cost:

- For batch pipelines → optimize cost
- For real-time alerts → prioritize speed, even if costlier

Tips:

Always use STAR method (Situation, Task, Action, Result).

Show decision-making + collaboration skills.

Tech Interview Titans